

PRODUCT PLAN · VERSION 1.0

HVAC Finance Tracker

A Polar Air Heating & Cooling web app to track job quotes, payments, materials, payroll, and profit — with a live bank feed that auto-categorizes income and expenses and lets every entry be edited.

Company

Polar Air Heating & Cooling

Date

August 18, 2026

Audience

Owner + office staff

1. Overview

This is a greenfield Next.js app for Polar Air. Version 1 combines **job-based profit and loss** with a **live bank feed**. Transactions import automatically, receive an HVAC category suggestion, and remain fully editable by the owner and office staff. The workspace starts empty; the app is built and hosted on Vercel.

This is an **operations ledger**, not a full payroll processor (no Gusto or ADP) and not a tax-filing replacement for QuickBooks. Payroll is a cost category and optional per-job labor — not paycheck generation.

2. What version 1 does

- Jobs:** customer, quote amount, status, payments received, costs (materials / payroll / other), and profit calculated as collected minus costs.
- Bank:** connect Polar Air’s checking (and later other accounts) via Plaid Link. Transactions import on their own and stay in sync.
- Categorize:** each import gets a suggested HVAC category (materials, payroll, job income, fuel, and so on). Anyone logged in can change category, job, splits, notes, or ignore it. Original bank data is never overwritten.
- Users:** Clerk login for the owner plus office staff, all seeing the same Polar Air books. Invite staff from Clerk; no public sign-up.

3. Technology stack

Provision integrations first, then build the app around them. Do not skip the bootstrap order.

Need	Provider	Why
App + hosting	Next.js App Router on Vercel	Fits the existing Vercel setup
Database	Neon Postgres via Vercel Marketplace	Job and transaction source of truth; Drizzle ORM
Auth	Clerk via Vercel Marketplace	Invite office staff; protect all app routes
Bank sync	Plaid Transactions	Industry standard; Polar Air never sees bank passwords
UI	shadcn/ui + Tailwind	Tables, dialogs, and forms for the ledger and jobs
Optional later	Vercel AI Gateway	Suggest HVAC categories when vendor rules miss

Plaid is **not** on the Vercel Marketplace. Create a Plaid dashboard account (Sandbox first; Production requires Plaid’s approval). Access tokens are stored encrypted in Neon and are never sent to the browser.

Bootstrap order (do not skip): create-next-app → vercel link → install Neon + Clerk → vercel env pull → add PLAID_* keys → Drizzle migrate → dev.

4. How money flows

Bank activity is the cash source of truth. Jobs receive allocations from that feed, plus any manual costs or payments that have not yet matched a bank line.



After review, each line feeds job P&L and company P&L. Manual job costs or payments can also land on a job.

Company profit comes from categorized bank activity: income minus expenses. Transfers and owner draws are excluded.

Job profit comes from money allocated to that job: payments versus materials, payroll, and other costs. When a bank charge is categorized as Materials or Payroll and assigned to a job, it counts once — no double-entry.

5. Data model (Drizzle + Neon)

Core tables:

- bullet **customers** — name, phone, email, service address
- bullet **jobs** — job number, customer, status (quoted → paid), quotedAmount in cents, dates, notes
- bullet **job_costs** — job, type (materials | payroll | subcontractor | permit | other), amount, date, description, optional transactionId
- bullet **job_payments** — job, amount, date, method, optional transactionId
- bullet **categories** — name, kind (income | expense | transfer | ignore), optional default job-cost type. Seed HVAC defaults: Job Income, Materials, Payroll, Fuel, Insurance, Utilities, Equipment, Permits, Advertising, Owner Draw, Transfer, Uncategorized
- bullet **vendor_rules** — merchant contains “Ferguson” → Materials; “ADP” / “Gusto” / “payroll” → Payroll; user-editable
- bullet **plaid_items / plaid_accounts** — institution, encrypted access token, sync cursor, balances
- bullet **transactions** — Plaid id, date, merchant, original amount and original Plaid category (immutable), current category, job, notes, reviewStatus (needs_review | confirmed), user override flags
- bullet **transaction_splits** — one bank line split across jobs or categories (for example a Home Depot ticket covering two jobs)
- bullet **audit_log** — who changed what on a transaction (important with office staff)

Money is stored as **integer cents**. Job P&L is computed, not stored, so quote versus collected versus costs never drift.

6. Bank connection and categorization

Every imported line is editable. Original bank values stay on the record so staff can recategorize without losing the source.

- 1 Settings → **Connect bank** opens Plaid Link.
- 2 Server exchanges the public_token, saves the item and accounts, and runs **/transactions/sync** (cursor-based; not the old **/transactions/get**).
- 3 Webhook **SYNC_UPDATES_AVAILABLE** (plus a daily Vercel cron as backup) pulls added, modified, and removed transactions.
- 4 For each new row: apply vendor rules → else map Plaid personal_finance_category → else Uncategorized + needs_review.

- 5 **Inbox** lists unreviewed items. Click any row to change category, assign or split to jobs, add notes, mark transfer/ignore, or confirm. Edits never mutate Plaid originals.
- 6 Amount stays the bank amount by default. An explicit override exists for rare corrections, with the original preserved.

Pending transactions update in place when they post (Plaid modified/removed events).

7. App screens

- Dashboard** — year-to-date collected, materials, payroll, other expenses, cash profit; jobs quoted vs collected; uncategorized count
- Jobs** — list and filters; job page with quote, paid, remaining, cost breakdown, profit, and linked bank lines
- Transactions** — inbox and full ledger; inline edit; split dialog
- Reports** — by category and by job for a date range
- Settings** — bank connection and sync, categories, vendor rules, invite staff (Clerk)

Visual direction: dark, operational admin UI (Geist + shadcn), Polar Air branding with a cool ice accent — not a marketing site.

8. Auth and access

Clerk Organizations: one organization named **Polar Air**. Sign-up is closed except Clerk invitations. All /dashboard and /api routes sit behind clerkMiddleware and auth.protect().

Role	Who	Access
Owner	Business owner	Everything, including disconnecting the bank
Office	Office staff	Jobs, transactions, categories, and rules. Cannot disconnect the bank or delete the organization.

9. Implementation sequence

- 1 Scaffold Next.js + shadcn; provision Neon and Clerk on the linked Vercel project; Drizzle schema plus seed categories and rules.
- 2 Auth shell: sign-in, invite, protected layout, Polar Air navigation.
- 3 Jobs CRUD and job P&L (quote, payments, costs) so the books work before the bank is live.
- 4 Plaid Link, sync, webhook, encrypted tokens; transaction inbox with full edit, split, and job assign.
- 5 Dashboard and reports; vendor-rule editor; audit log.
- 6 Deploy to Vercel; Production Plaid after Sandbox works.

Build checklist

#	Workstream	Status
1	Scaffold Next.js, link Vercel, provision Neon + Clerk, Drizzle schema, HVAC category seed	Pending
2	Clerk org auth, Polar Air app shell, owner vs office roles, invite-only access	Pending

#	Workstream	Status
3	Customers, jobs, quote / payments / costs, per-job profit UI	Pending
4	Plaid Link, encrypted tokens, transactions/sync, webhooks, cron backup	Pending
5	Rules + Plaid mapping, review inbox, full edit / split / job assign, audit log	Pending
6	Company dashboard and category / job reports; deploy	Pending

10. What you will need to do once (outside the code)

- bullet Approve Neon and Clerk Marketplace installs when prompted.
- bullet Create a Plaid account and paste PLAID_CLIENT_ID / PLAID_SECRET / PLAID_ENV into Vercel env.
- bullet Invite office staff in Clerk after the first deploy.
- bullet For live bank data: complete Plaid Production verification. Sandbox uses fake banks only.

11. Out of version 1 (on purpose)

- bullet Customer-facing quotes, invoices, or portals
- bullet Direct payroll processing (ADP / Gusto)
- bullet Multi-company or technician-only logins
- bullet Tax packages or 1099 generation
- bullet Receipt OCR / photo attachments (easy add later via Vercel Blob)

Polar Air Heating & Cooling · HVAC Finance Tracker plan · August 18, 2026. This document describes the agreed v1 product: job-based P&L plus a live, fully editable bank feed.